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Original Article

Individual aggression, but not winner–loser effects, predicts social rank in male domestic fowl

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Many factors can affect the probability for an individual to obtain a high social rank, including size, weaponry, and behavioral attributes such as aggression. Recent experiences of winning or losing can also affect the chances of winning future contests, commonly referred to as "winner–loser effects". Individuals often differ in behavior in a consistent way, including in aggression, thereby showing differences in personality. However, the relative importance of recent experience and aspects of personality in determining rank, as well as the extent to which winning or losing affects aggression, has rarely been studied. Here, we investigate these questions using male domestic fowl. We matched males for body size, comb size, and aggression in pair-wise duels to: 1) study the effect of contest outcome on aggression and 2) compare the effect of individual aggression and contest experience on future social status in small groups. We found that aggression was a highly repeatable personality trait and that aggression increased after winning and decreased after losing. Nevertheless, such winner–loser effects were not enough to increase the odds of becoming dominant in a small group. Instead, aggressiveness measured prior to a contest experience best predicted future rank. Boldness and exploration did not predict rank and of the 2, only boldness was positively correlated with aggressiveness. We conclude that for male domestic fowl in contests among phenotypically matched contestants, aggressiveness is more important for obtaining high rank than winner–loser effects, or other aspects of personality.

Key words: **behavioral syndrome, *Gallus gallus domesticus*, personality, resource holding potential, social dominance status, social hierarchies.**

INTRODUCTION

Dominance relationships characterize social structure in many animal species. Being in a dominant social position increases access to important resources such as mating partners (Andersson 1994; Clutton-Brock and Huchard 2013). Social position can therefore influence reproductive success and can be a significant contributor to individual variation in fitness. Competition for social dominance commonly follows stereotypic escalating patterns from aggressive displays to physical attacks, ending at any stage when one of the competitors withdraws (e.g., Collias 1943; Parker 1974).

The outcome of a contest has been shown to affect future contests, so that winners increase their probability of winning a subsequent contest, whereas losers tend to keep on losing, also against new opponents (reviewed in Hsu et al. 2006; Rutte et al. 2006). Such "winner–loser effects" have been observed in several species (e.g., Ginsburg and Allee 1942; Bakker and Sevenster 1983; Chase et al. 1994; Whitehouse 1997; Hsu and Wolf 1999; Kasumovic et al. 2010; Trannoy et al. 2016). It has been suggested that winning and losing contributes to a male's information of his own fighting capacity relative to others in the population (Whitehouse 1997; Hsu and Wolf 2001). This self-assessment can then be used to optimize decisions in future interactions to avoid costly fights (Rutte et al. 2006). Changes in specific elements of agonistic behavior may be a key to winner–loser effects (Hsu and Wolf 2001; Kar et al. 2016). For example, the probability of initiating and escalating a new fight increases in winners of the mangrove killifish (*Rivulus marmoratus*), whereas losers are more prone to avoid further contests (Hsu and Wolf 2001). Nevertheless, few studies have aimed to describe changes in agonistic behavior outside the contest context in which the winner or loser effect is measured (McEvoy et al. 2013).

The ability to win a contest varies among individuals and depends on several attributes. Physical attributes such as body size, weapon size, strength, and endurance have been much studied and are often positively linked to dominance (Andersson 1994; Arnott and Elwood 2009; Clutton-Brock and Huchard 2013). The role of behavioral traits has received much less attention, except the often observed link between aggression and the chance of winning a duel (e.g., Wilson et al. 2011; Briffa et al. 2015). Aggressiveness, the individual reaction in agonistic intraspecific interactions (Réale et al. 2007), can be a plastic trait and has often been studied from

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a context-dependent perspective, focusing on circumstances where animals chose to continue fighting or withdraw (Arnott and Elwood 2009). However, recent empirical studies have shown that aggressiveness also differs substantially among individuals and that these differences can be consistent across time and/or context (Réale et al. 2007; e.g., Wilson et al. 2011; McEvoy et al. 2013; Favati, Leimar, Løvlie 2014; Camerlink et al. 2015) and can therefore be considered a personality trait (Dall et al. 2004; Sih et al. 2004). Some studies have shown that personality traits can predict the outcome of agonistic interactions. In general, more aggressive, active, explorative, and bold individuals have higher chances of obtaining dominant positions (e.g., Riechert and Hedrick 1993; Verbeek et al. 1996; Colleter and Brown 2011; Dahlbom et al. 2011; Favati, Leimar, Løvlie 2014) and these traits sometimes covary to form "behavioral syndromes" (i.e., behaviors correlated across time and/or context, Sih et al. 2004).

The predictive value of personality traits for contest outcome has mostly been explored in single pair-wise interactions. Such a set up excludes the presumably more common situation in which an individual has to meet several consecutive opponents to establish its position within a group, as dominance hierarchies form or change in group-living animals. There are 2 main hypotheses regarding how an individual gains its position within dominance hierarchies in groups of animals, either through prior attributes, or through self-organization/social dynamics. Differences in individual attributes such as size, age or aggressiveness may determine dominance rank ("prior attribute hypothesis"), which can be mirrored through the outcome of pair-wise interactions (Chase et al. 2002; Chase and Seitz 2011). However, prior attributes and rank obtained in isolated pairs are rarely fully predictive of rank in groups and it has been suggested that additional processes such as winner, loser, and bystander effects are needed to form (linear) hierarchies ("self-organization hypothesis" or "social dynamics hypothesis," Chase et al. 2002; Dugatkin and Earley 2003; Chase and Seitz 2011; Dey and Quinn 2014). Winner–loser effects have nevertheless mainly been studied in pair-wise interactions and the potential importance of winner–loser effects on obtained rank in groups is therefore unclear. Furthermore, these hypotheses are not mutually exclusive (Chase and Seitz 2011; Dey and Quinn 2014) but aggressiveness and winner–loser effects have hitherto almost exclusively been investigated separately within this framework. The aim of this study is therefore to investigate if recent experiences of winning or losing affect aggression and to compare the relative importance of recent contest experience and personality for the outcome of establishment of social dominance in small groups.

By arranging duels with males of domestic fowl (*Gallus gallus domesticus*) matched according to aggressiveness, we produced winners and losers that did not differ in individual aggressiveness and that had received the same social experimental treatment, except for the winner–loser experience. This design offers an opportunity to study the separate effects of winning, losing, and individual aggressiveness. We used this approach in 2 experiments investigating: 1) the effect of winning and losing on aggression and 2) the relative effect of contest outcome and individual aggressiveness on future social rank in small groups of male domestic fowl.

The domestic fowl (*Gallus gallus domesticus*) is a well-studied model species for studies of dominance and aggression (Chase 1980). Both sexes readily form distinct (sex-specific) dominance relationships through dyadic agonistic interactions and dominance positively affects reproductive success (females, Collias et al. 1994; male matings, Collias and Collias 1996). Size of body and ornament (comb), as well as age and personality, can substantially, but not fully, predict the outcome of duels (e.g., Ligon et al. 1990; Zuk and Johnsen 2000; Parker et al. 2002; Favati, Leimar, Løvlie 2014). Aggression displayed early during an agonistic interaction (i.e., aggressive posture obtained when contestants first meet and the initiator of the first attack) can foretell the outcome (females, Collias 1943; Foreman and Allee 1959; males, Favati, Leimar, Løvlie 2014) but using the same interaction to measure both aggressiveness and determining dominance outcomes suffers from circularity. Therefore, an independent estimation of individual aggressiveness is needed to more accurately explore to what extent intrinsic aggressiveness may predict dominance. Dominance hierarchies are relatively stable over time, yet individuals occasionally disperse and enter new groups, facing a situation in which they have to challenge several flock members (Collias and Collias 1996). Winner–loser effects have been demonstrated in hens (Chase 1980; Martin et al. 1997; Kim and Zuk 2000) but have to our knowledge not been specifically examined in males. Further, recent changes in social position affect nonsocial behavior such as activity and vigilance (Favati, Leimar, Radesäter, et al. 2014); however effects on aggressiveness remain to be investigated.

METHODS

Study population and animal housing

The study was conducted during the breeding season of 2012 (June to August) at Tovetorp Research station, Stockholm University, using 54 males of the old Swedish game breed of fowl "Swedish bantam" (*Gallus gallus domesticus*, "Gammalsvensk dvärghöna" in Swedish). Birds were housed in 6 aviaries (6 × 10 m) in mixed-sex, mixed-aged groups (1–13 years, 15–20 birds per group, sex ratio ~1:1) with ad libitum access to food and water. The aviaries were floored with sand and fitted with nest boxes, perches, and fresh bushes and branches. Before behavioral assays, individual males were isolated from other males and housed together with 2 females for 2–3 days in smaller aviaries (approximately 3 × 3 m) under otherwise the same conditions as described above. This was done to reduce the effect that previous social rank can have on behavior (e.g., Cornwallis and Birkhead 2008; Favati, Leimar, Radesäter, et al. 2014).

We here study social hierarchy formation among unfamiliar male domestic fowl. These contests may be aggressive and include physical interactions. All contests were therefore closely monitored by live observation and the observers were ready to interrupt if one of the males was injured or showed signs of fatigue. However, this was not needed in any case. The study fulfilled the ethical requirements in Sweden and was approved by Linköping Ethical Committee (ethical permit number 60-10).

General experimental setup

To investigate the potential relationship between personality and the outcome of later agonistic interactions, we conducted 3 personality assays: a novel arena test (n = 40), a novel object test (n = 40) and an aggression test (n = 39 in experiment A, and n = 26 in experiment B), and 2 winner–loser experiments: "experiment A" (n = 39) and "experiment B" (n = 26). Figure 1 gives a schematic overview of the experimental setup and detailed description of each assay follows below. The novel arena and novel object tests were carried out first, in random order with at least 4 h between tests. Thereafter experiments A and B were performed, in random

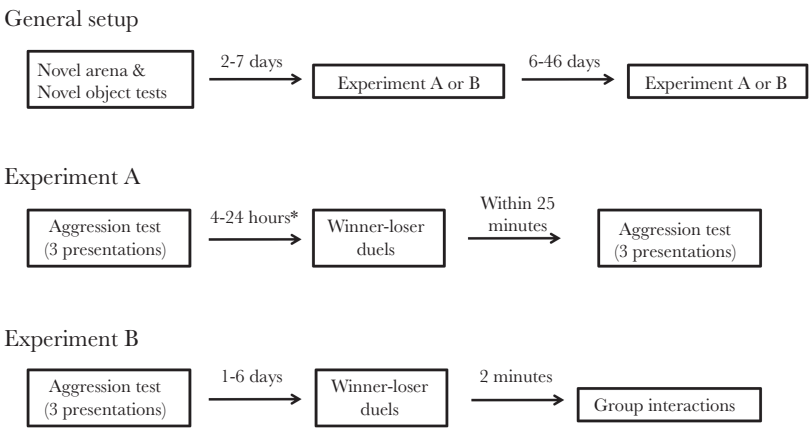

Figure 1
Schematic overview of the experimental setup. All males were exposed to a novel object test and a novel arena test before the main experiments (10–28 days before experiment B). The order of experiment A and B was randomized. Each aggression test consisted of 3 trials (see Methods for details). *Except 4 cases, where there were 14 days between tests.

order and 6–46 days apart. Each of these experiments started with scoring male aggressiveness. The aggression scores obtained were used to match males for aggressiveness in the following winner–loser duels. An additional aggression test was conducted at the end of experiment A in order to investigate the effect of winning and losing on individual aggression. A subset of males (n = 21) was scored in the novel arena and novel object tests a second time 1–2 months after the first test in order to investigate the consistency of personality traits measured in these assays.

Personality assays

Scoring aggression

Individual aggression was scored by observing the focal male's behavior while presenting a restrained male, using the method described by Favati, Leimar, Løvlie 2014). In short, an unfamiliar male (that had not been housed together for at least 2 weeks; $n_{\text{presented males}}$ = 23, whereof 15 were reused focal males from experiment A, see below) was manually restrained in the hands of the observer and presented on the ground to the focal male. The presented male had a smaller or equally sized comb compared to the focal male, this with the aim to reduce any potentially intimidating effects of a large-combed opponent. Comb size was estimated by measuring the length of the comb to the nearest mm using a digital caliper. Aggressive behavior by the focal male was observed for 1 min or until he initiated an attack on the restrained male, whichever came first. The male's level of aggression was classified along a 7-grade scale (an expanded version of the 4-grade scale used in Favati, Leimar, Løvlie 2014) ranging from avoiding the presented male (score 0) to being very aggressive (score 6; Table 1). At every occasion of scoring aggression, 3 different intruders were presented to achieve a robust aggression measurement.

Novel arena test

All males were individually exposed to a novel arena for 10 min, using the method described in Favati, Leimar, Løvlie 2014). The arena was an oval shaped fenced area (11.5 × 10 m) in a deciduous forest, divided into 32 roughly even-sized subareas. Based on previous personality assaying in the fowl (Favati, Leimar, Løvlie 2014; Favati, Leimar, Radesäter, et al. 2014; Favati et al. 2016; Zidar et al. 2017), the following behaviors were recorded. "Vigilance" was measured as the proportion of observations (recorded every 15 s) a male had his head above shoulder height (as opposed to pecking on the ground or preening). "Exploration" was measured as a combination of 3 variables: the latency to visit 5 subareas, the total number of subarea changes made, and the number of new subareas visited, and these measures were z-standardized and combined (averaged) to generate one combined "exploration" measure. In forming the combined measure, latency to explore 5 subareas was first multiplied by −1 to obtain a variable where a higher value describes males that were more explorative.

Table 1
Classification of aggression in male domestic fowl in response to presentation of a manually restrained, live opponent

Score	Description
0	Walks away from the other male
1	Straight body posture, no avoidance or attraction to the other male
2	Aggressive posture after > 30 s
3	Aggressive posture within 30 s
4	Aggressive posture + aggressive ground pecking within 30 s and/or approach opponent < 50 cm within 60 s
5	Aggressive posture + approach opponent < 50 cm within 30 s
6	Approach opponent < 50 cm within 10 s

The aggression scores range from 0 to 6 (6 being highest). Males received score 3 when they adopted an aggressive posture within 30 s (one or more of tilted and/or crouched posture, dropped wing, raised hackles, Collias 1943, Kruijt 1964). Males received a lower score if they took longer than 30 s to show aggression or if no aggression was shown, and a higher score if they in addition performed the aggressive display ground pecking (Kruijt 1964) or approached the presented male.

Novel object test

A novel object test was performed to score individual boldness (sensu Favati, Leimar, Løvlie 2014; Favati et al. 2016. The female companions of the focal male were removed from the home cage, and a mealworm was placed on a feeding plate in a corner of the cage to attract the male to the plate. When the male had eaten the mealworm, he was gently herded to the opposite corner of the cage. A new mealworm was placed on the plate, and a novel object (a black toy rat, ca 8 cm long) was placed 10 cm behind the plate. The observer backed off, and immediately started a stopwatch. The latency for the male to approach the plate (within approximately a

body length, ca 40 cm, from the plate) from the opposite corner of the cage (~3 meters) was recorded as a measure of "boldness," that is, a low score indicates a bolder male. Males that did not approach the plate (n = 5) were assigned the maximum latency of 300 s.

Generating winner–loser experiences

There are 2 principal ways to experimentally produce winners and losers. One can pair focal individuals with individuals that are either likely to win (e.g., a larger resident) or lose (e.g., a smaller intruder), to produce a winner or loser experience (e.g., Hsu et al. 2006; Kloke et al. 2011). The drawback of this method is that winners and losers do not receive the same experimental treatment, making it difficult to directly compare winner and loser effects. Alternatively, 2 individuals can be paired to settle a fight, resulting in "self-selected" winners and losers. Such winners and losers are likely to differ in aggressiveness or other traits important for dominance and these traits may thus be confounded with potential winner–loser effects (Chase 1994). It has therefore been suggested that a contest with a phenotypically matched opponent provides more information of an individual's own perceived fighting ability (Beaugrand and Goulet 2000). We have chosen the latter method and also match contestants for aggressiveness in order to further distinguish the winner or loser experience from individual qualities.

To induce either a winner or loser experience in the 2 experiments, 2 males were simultaneously placed in an aviary unfamiliar to both. Males within a pair were matched for aggression score (mean score of 3 presentations with different intruders, maximum difference 2 grades), comb size (comb length, maximum ~10% difference), body weight (measured to the nearest g, maximum ~10% difference), and age (1–8 years, maximum difference 2 years). Comb size and weight were measured the week before each of the 2 experiments. Five consecutive avoidances from a male defined the loser of the duel and the other as the winner (Favati, Leimar, Radesäter, et al. 2014). A winner and a loser were identified within 30 min in all duels and duels were often resolved within a few minutes. No males were injured more than minor bleeding from the comb during these trials.

Experiment A: the effect of winning or losing on aggressiveness

To study the effect of winning or losing on subsequent male aggressive behavior, aggression was scored on 2 occasions: before and after a duel with another male (performed as described above), which caused males to either gain a winning experience, or a losing experience (Figure 1). Forty males were used in experiment A. One male became sick during the isolation period and could not participate in the experiment. To obtain even pairs for the winner–loser duels, he was replaced by another male, that was not included in any analyses, resulting in n = 39 ($n_{winners}$ = 20, n_{losers} = 19). Aggression was scored 4–24 h before the duel. Approximately 2 min after the winner and loser were identified in a duel, the males were separately transferred to new aviaries, and aggression was again scored within approximately 25 min.

Experiment B: the effect of winning or losing on future social rank

To compare the potential effects of individual aggression and recent experience of winning or losing on future social rank in a small group of males, aggression was scored before a winner or loser experience was induced by a duel as described above (Figure 1). Two minutes after identifying the winner and the loser of a duel, each male was separately introduced to a small group consisting of 2 randomly chosen weight-matched, unfamiliar males, resulting in a group of 3 males. To investigate potential resident–intruder effects (Leimar and Enquist 1984), both males of a pair were either introduced as the intruder to their respective groups (n = 6), as residents (by being introduced to the original cages where the focal males had been isolated during the experiment, n = 6), or simultaneously to the group of males in an aviary unfamiliar to both the focal male and the group (n = 14). These groups were observed until the dominant male could be identified using the same criteria as described above for duels. If the dominant male could not be identified within 2 h (due to low rate of interactions), observations continued the next day until a clear hierarchy was observed. The dominant male was clearly identified in all groups. Of the 40 males used in experiment A, we were able to match 13 pairs of males for the winner–loser duels, with 2 opponents each for experiment B, resulting in 26 focal males used in the analyses. Four of the males in experiment A were not possible to match and were left unused in experiment B. A total of 21 different opponent groups were formed using 29 males, whereof 20 were reused from experiment A and/or B, and 9 that were only used as opponents.

A grand total of 54 males were used: 40 males were used as focal animals in experiment A and/or B and 14 additional males were used as opponents in experiment B or as restrained intruders in the aggression tests.

Statistical analyses

Consistency and correlation of personality traits

Spearman rank correlations were used to investigate rank-order consistency in aggression (aggression scored before experiment A versus aggression scored before experiment B, in chronological order for each individual), vigilance, exploration, and boldness. We also computed intraclass correlations coefficients (ICC) for the same comparisons using the R statistical software package rptR (Nakagawa et al. 2010; Schielzeth et al. 2016). Histograms and Box-Cox diagrams were assessed to determine the distribution of data. Aggression score, vigilance, and exploration were normally distributed and therefore linear mixed models with Gaussian distributions were fitted by REML (1000 bootstraps, 1000 permutations) to estimate repeatability. Boldness was characterized by a bimodal distribution and was therefore converted to a binary variable (0 for males with a latency to approach ≥ 60 s and 1 for males with a latency < 60 s) that was used in all further analyses. Repeatability was estimated by fitting a generalized linear mixed model with binomial distribution and logit link. To investigate the relationships among personality traits (the first mean aggression score, vigilance, exploration, and boldness) we performed Spearman rank correlations.

To control for a successful matching of males within pairs, the difference in mean aggression score before the pair-wise duels, comb size, weight and age between the winner and loser were tested using one-sample t-tests (i.e., equivalent to a paired t-test). The data met the assumptions for parametric tests. To check if the traits males were matched for body weight, comb size, and aggression could still influence the outcome of duels, we performed a logistic regression model with outcome of the duel as response variable and the matched traits as fixed factors. Only body weight differed between future winners and losers in experiment B (see Results for details) and was therefore included in further analyses (see below).

Experiment A: the effect of winning or losing on aggression

To investigate the effect of winning and losing on aggression, we used a linear mixed model with aggression score as response variable. The outcome of the duel (winner versus loser), the scoring time (before or after the duel, in total 6 scores per male) and their interaction were set as explanatory factors, and identity of the male was set as a random factor. Visual inspection of residual plots was performed to confirm that the aggression scores met the assumptions for a linear regression.

Experiment B: the effect of winning or losing on future social rank

The outcomes of group interactions were investigated by 3 separate logistic regression models with the outcome of the group interaction (dominant vs. lower rank) as response variable. The first model initially contained recent contest experience (winner/loser), mean aggression score before the duel (0–6), the contest experience by aggression score interaction term, and body weight as explanatory variables. All numerical explanatory variables were *z*-standardized in order to facilitate comparisons of effects. However, this model had problems with complete separation (for the subset of losers, as a function of the aggression score), which caused the glmer fitting to become unreliable. As a workaround, we used a Bayesian MCMC analysis to estimate a regression model (R package rstanarm, Stan Development Team 2016). This approach can handle complete separation using weak priors for the regression coefficients (Gelman et al. 2008). We used the default priors of the intercept and coefficients for the stan_glm function (Gaussian priors with location = 0, scale = 10 for the intercept and location = 0, scale = 2.5 for the coefficients). The effective sample size aimed for was 4000. The winner/loser by aggression score interaction term turned out to be nonsignificant and was therefore dropped from the model. Weight was also not statistically significant and was also dropped from the model. The final model was thus: outcome ~ aggression score + recent contest experience. To evaluate whether personality traits other than aggression could predict the outcome of dominance in groups while also minimizing the risk of over-parameterization, we ran a generalized linear model (binomial error distribution and logit link) with vigilance, exploration, and boldness (all measured before experiment B) as predictors. Similarly, we ran a model with mode of introduction to the group (intruder, resident, mixed) and body weight as explanatory variables.

All analyses were performed using the statistical software R, version 3.3.1 (R Core Team 2016) and for the mixed model fitting, we used the lme4 package (Bates et al. 2015) and the rstanarm package (Stan Development Team 2016).

RESULTS

Consistency and intercorrelation of personality traits

Aggressiveness was highly rank-order consistent over time (n = 25, Spearman, r = 0.83, P < 0.0001, ICC = 0.79, 95% CI: 0.59, 0.90, Figure 2). Vigilance was consistent between test occasions (n = 21, r = 0.39, P = 0.08, ICC = 0.45, 95% CI: 0.047, 0.73), while exploration (n = 21, r = 0.33, P = 0.15, ICC = 0.27, 95% CI: 0, 0.62) and boldness (n = 21, ICC = 0.25, 95% CI 0.0, 0.67) varied more. More aggressive males tended to be less vigilant than less aggressive males (n = 41, r = −0.10, P = 0.054) but aggression did not correlate with exploration (n = 41, r = −0.10, P = 0.55). More aggressive males were bolder in the novel object test (i.e., had a shorter latency to approach the object, n = 41, r = −0.35, P = 0.03).

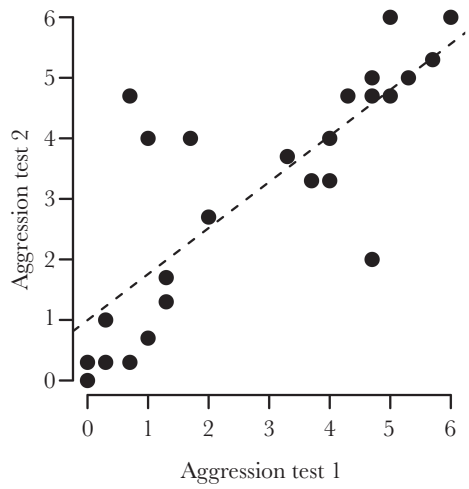

Figure 2
Aggression scores of male domestic fowl scored at 2 different test occasions. Males were consistent in their aggression score when presented to an unknown, restrained opponent, with approximately 1 month between tests. The scores are the mean value of 3 presentations at each test occasion.

Experiment A: the effect of winning or losing on aggression

Phenotypic matching of males was successful and there was no difference between future winners and losers within pairs (n = 19) of males in aggressiveness (mean difference: 0.1 grades ± 0.3 SE, t = 0.46, df = 18, P = 0.65), comb size (mean difference: 1.7 mm ± 0.3 SE, t = 1.34, df = 18, P = 0.19), body weight (mean difference: 2.0 g ± 26.8 SE, t = 0.073, df = 18, P = 0.94), or age (mean difference: −0.2 years ± 0.3 SE, t = −0.70, df = 18, P = 0.49). Also, none of these traits could predict the outcome of the winner–loser duels (logistic regression, n = 39, all P > 0.05, for further details see Supplementary Table S1). For the main question of the effect of winning or losing on aggression, an analysis of the linear mixed model

$$\text{Aggression score} \sim \text{Duel outcome} * \text{Scoring time} + (1 \mid \text{Male Identity}),$$

presented in Table 2, showed that there was a statistically significant interaction between the duel outcome (lose/win) and scoring time (before/after duel). This means that winners and losers differed in how their aggression score changed from before to after the duel. Examining the changes in aggression scores separately for winners and losers, by computing profile confidence intervals from the mixed model fitting, showed that winners increased their aggression score (n = 20, estimated change = 1.100, 95% CI: 0.696, 1.504), while losers decreased in score, but to a lesser extent (n = 19, estimated change = −0.421, 95% CI: −0.835, −0.007). The average aggression scores before and after the duel, for winners and losers, are depicted in Figure 3.

Experiment B: the effect of winning or losing on future social rank

Males were successfully matched within duel pairs (n = 13) also before experiment B, except for body weight; future losers tended to be heavier (winner–loser differences: aggressiveness, mean

Table 2

Model output for a linear mixed model of aggression scores in male domestic fowl, in experiment A

Effect	Estimate (± 95% CI)	SE	*P*
Intercept	4.105 (3.314, 4.896)	0.405	
Duel outcome	0.228 (−0.876, 1.332)	0.565	0.070
Scoring time	−0.421 (−0.835, −0.007)	0.211	0.015
Outcome × time	1.521 (0.942, 2.099)	0.295	<0.001

The response variable was the aggression score and there were 3 scoring occasions before and 3 after the duel, which thus are repeated measures for individual males (n = 39 males). The intercept in the model is the score for losers before the duel, the effect of duel outcome is the winner–loser difference before the duel, the effect of scoring time is the before–after difference for losers and the interaction is the deviation from additivity for the before–after difference for winners. The confidence intervals given are 95% profile confidence intervals computed by the confint function from the lmer model fitting.

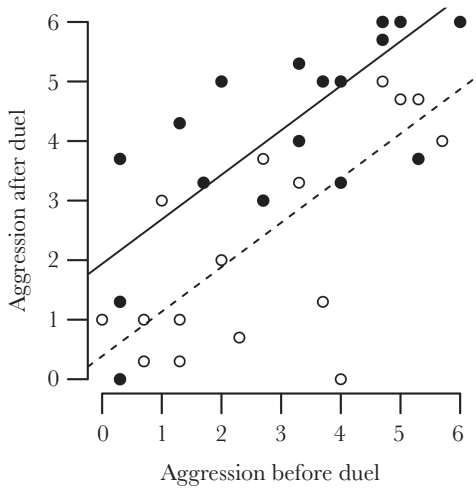

Figure 3

Aggression scores of male domestic fowl before and after a duel. Male domestic fowl that had recently won a duel (filled dots, solid line) had significantly higher aggression score compared to males that had lost a duel (open dots, dashed line).

difference: −0.3 grades ± 0.3 SE, t = −0.75, df = 12, P = 0.47; comb size, mean difference: −2.6 mm ± 1.2 SE, t = −1.53, df = 12, P = 0.15; body weight, mean difference: −82.8 g ± 17.5 SE, t = −3.42, df = 12, P = 0.0051; age, mean difference: −0.5 years ± 0.3 SE, t = −1.20, df = 12, P = 0.25). In experiment B, only individual aggressiveness measured before duels predicted the outcome of group interactions (n = 26): more aggressive males became dominant more often (Table 3, Figure 4a). Recent contest experience (winner/loser of a duel) did not have a statistically significant effect on the outcome of group interactions (Table 3, Figure 4b). None of the personality traits measured in novel arena and novel object tests could predict the outcome of group interactions (Table 3) and neither could mode of introduction nor body weight (Table 3).

DISCUSSION

Our study shows that aggressiveness increased in male domestic fowl that had recently won a fight, while males that lost decreased in aggressiveness but to a lesser degree. However, when a male that had recently won or lost a fight was introduced into a small group of unfamiliar males, the recent social experience did not affect the social position he obtained. Instead, individual aggressiveness prior to the winner–loser experience predicted whether or not he would obtain a dominant position. Our results therefore suggest that personality, or at least aggression, has a stronger effect on the competition for social dominance than recent social experience.

Table 3

Fit of logistic regression models of the probability of becoming dominant in a group of male domestic fowl, in experiment B

Model	Model term	Estimate	SE	*P*
1	Experience	0.583 (−1.173, 2.428)	0.179	
	Aggression score	1.339 (0.409, 2.437)	0.101	
2	Vigilance	−0.124	0.452	0.783
	Exploration	−0.420	0.436	0.336
	Boldness	−0.168	0.434	0.700
3	Mode of introduction			
	Intruder	0.987	1.027	0.336
	Resident	1.890	1.227	0.123
	Body weight	−0.028	0.456	0.952

The outcome of 3 separate models is presented (1–3). The first model is a Bayesian analysis. Estimates are given ± 95% CI. Aggression score was the only parameter that significantly predicted the outcome in group interactions (model 1). The baseline for "Experience" in the analysis was the loser experience. Therefore, the positive estimate means that winners had slightly higher chances of becoming dominant in the following group interaction compared to losers, however not significantly so, because the 95% confidence interval (CI) includes 0. All estimates from the logistic regression model fittings are given on the logit scale. The baseline for mode of introduction is simultaneous introduction.

Recent experience of winning or losing can have an impact on conflict resolution. The proximate causes of winner–loser effects are not known in full detail but recent studies show that changes in social position are associated with changes in neurogenomic states (e.g., through DNA methylation in the brain, Cardoso et al. 2015) and hormonal responses that can govern agonistic behavior (e.g., Teles and Oliveira 2016). The increased chance for winners to continue to win, and losers to continue to lose, is likely to be mediated through behavioral changes following the first win or loss, although the precise behavioral mechanisms are rarely studied. A meta-analysis showed that losing experiences decrease the probability to initiate new contests, whereas winning increases an individual's willingness to escalate a contest (Hsu et al. 2006). The winner and loser effects may thus act through different behavioral changes. In the current study, contest experience affected behavior such that aggressiveness increased in winners and decreased in losers. Simultaneously, aggressiveness was highly repeatable across contests and time, confirming that aggressiveness is a consistent personality trait, in line with previous studies of various species (e.g., Franck and Ribowski 1987; Réale et al. 2007; Wilson et al. 2011; McEvoy et al. 2013; Favati, Leimar, Løvlie 2014; Camerlink et al. 2015). This demonstrates that behavioral plasticity and consistency need not to be mutually exclusive (e.g., Briffa et al. 2008; Favati, Leimar, Radesäter, et al. 2014), although contests might interrupt consistency of personality traits (Courtene-Jones and Briffa 2014). The results are similar to findings in fish, where aggressiveness has been observed to be both intraindividually consistent and change with contest experience (Franck and Ribowski 1987; Karino and Someya 2007).

In our winner–loser experiments, males were matched for morphological traits (body weight, comb size), age and aggressiveness, all being traits that have previously been shown to affect the outcome of agonistic interactions in the species (e.g., Ligon et al. 1990; Zuk and Johnsen 2000; Parker et al. 2002; Favati, Leimar, Løvlie 2014). Our setup created an artificial situation with males being

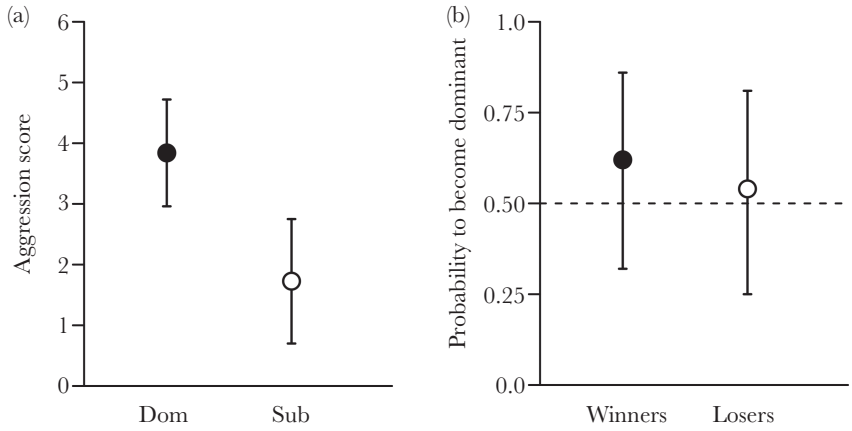

Figure 4

Predictors of dominance rank in small groups of male domestic fowl. (a) Aggression scores (mean ± 95% CI) of future dominant and subdominant males. Aggressive individuals more often obtained the dominant position in groups ("Dom," filled dot), whereas less aggressive males more often obtained a subordinate position ("Sub," empty dot). (b) Probability of becoming dominant in small groups (experiment B, see Methods for details). Males with recent experience of winning a duel ("Winners," filled dot) did not have significantly higher chances of obtaining a dominant position in subsequent group interactions, compared to males with recent experience of losing ("Losers," empty dot). Bars show 95% confidence intervals. The dotted line marks 50% chance to win or lose, that is, the expected probability by chance.

fairly closely matched for several traits, which had the advantage of allowing us to study the effects of winning and losing on individual aggressiveness with minimal interference of these traits. In a more natural situation, males will vary in all traits, and which one is the stronger predictor of post-conflict behavior and outcome of future contests warrants further studies.

The process of acquiring a social position within a group may be different from settling dominance within a pair. When individuals meet several opponents simultaneously, not only differences in attributes but also additional processes like winner–loser and bystander effects can come into play (Chase et al. 2002; Dugatkin and Earley 2003; Chase and Seitz 2011; Dey and Quinn 2014). However, both personality as a prior attribute and winner–loser effects have mainly been studied in isolated duels. A few studies show that winner–loser effects also apply in small groups (e.g., Cloutier et al. 1996; Dugatkin and Druen 2004) but winner–loser effects have in these cases not been disentangled from intrinsic fighting capacity or aggressiveness. The impact of morphological traits on contest success has been compared with the importance of recent experience (e.g., body size, Beaugrand and Cotnoir 1996; weight, Martin et al. 1997), whereas individual behavioral characteristics have not been compared with experience to the same extent.

Our study is one of the first designed to disentangle these effects. Our experimental setup where males were matched for aggressiveness in pair-wise duels effectively separated innate aggressiveness from contest experience. Observed differences in aggressiveness between winners and losers could therefore be assigned to winning or losing per se, with reduced interference of individual capacity or aggression. In the group context experiment (experiment B), the competition a male met in the form of a small group of males represents a fairly natural situation where an individual needs to face several opponents of varying quality in a row. Although we found that aggressiveness was affected by contest experience (in experiment A), previous wins or losses had no statistically significant effect on future dominance rank in small groups of males (in experiment B). A potential interference between aggressiveness and winner–loser effects is rarely controlled for, and it is thus possible that some of the winner–loser effects observed in other studies are confounded by individual differences in innate aggressiveness. Alternatively, winner–loser effects might not appear unless strengthened by individual recognition (Trannoy et al. 2016), signals of status (Pagel and Dawkins 1997), or familiarity with the site of previous defeats (Cloutier et al. 1995). We showed that aggressiveness predicted the outcome of social interactions in a small group, in line with previous results in pair-wise duels in the same population of fowl (Favati, Leimar, Løvlie 2014). Individual characteristics thereby predict future rank not only in pairs but also in small groups of male domestic fowl, in accordance with the "prior attributes hypothesis" (Chase et al. 2002). We did not observe strong winner–loser effects, which is similar to what was found in a study of green anoles (*Anolis carolinensis*), where aggressiveness in the form of participation in escalated fights best predicted winning secondary contests (Garcia et al. 2012). However, individual aggression is not always predictive of future contest outcomes (Bakker and Sevenster 1983; McEvoy et al. 2013; Camerlink et al. 2015), which questions aggressiveness as a general honest signal of fighting ability. Different personality types are likely to differ in their response to winning or losing a contest (Briffa et al. 2015) and our setup had the potential to test this idea. However, the absence of winner–loser effects on future rank in our study prevented further exploration of this idea.

Aggressiveness sometimes covaries with other personality traits. A commonly found behavioral syndrome is a positive correlation between aggressiveness and boldness (e.g., Huntingford 1976; Kortet and Hedrick 2007), or aggressiveness and exploration (e.g., Verbeek et al. 1999, Santostefano et al. 2016), although the association between aggressiveness and other personality traits is weak or even absent in some studies (e.g., Bell 2005; Chang et al, 2012; Favati, Leimar, Løvlie 2014). In the current study, aggressive individuals were bolder and less vigilant compared with less aggressive individuals, while we found no significant association between aggressiveness and exploration. The varying results have been suggested to depend on environmental factors like predator pressure (Bell and Sih 2007) but the mechanisms underlying correlations between traits are not yet well understood.

Although there seems to be a relationship between aggressiveness, boldness and vigilance in male fowl, neither boldness, neither vigilance, nor exploration could predict future social rank in our experimental group interactions. This is in contrast to an earlier

study of the same population of fowl, where fast explorers became dominant in single pair-wise duels (Favati, Leimar, Løvlie 2014). Similar patterns have been observed in great tits (*Parus major*) in which the relationship between exploration speed and dominance was positive in single pair-wise interactions but less clear in a group context where fast explorers ended up in either top or bottom rank positions (Verbeek et al. 1996, 1999). Further studies in varying social contexts are therefore needed to describe the role of personality on hierarchy formation of groups. Currently, aggressiveness appears to be the personality trait that best predicts future rank in male fowl.

To conclude, we have demonstrated that aggressiveness was boosted in winners and slightly reduced in losers. Nevertheless, the boosted aggressiveness of winners was not strong enough to induce a winner effect in a following group interaction in male domestic fowl and neither did the experience of losing influence future interactions. Instead, individual aggressiveness appeared to be the major predictor of the outcome of dominance in groups of males. Further studies testing various aspects of success and failure in contests may elucidate to what extent the impact of personality traits overrides winner–loser effects and the role of size of social groups.

SUPPLEMENTARY MATERIAL

Supplementary data are available at *Behavioral Ecology* online.

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Data accessibility: Analyses reported in this article can be reproduced using the data provided by Favati et al (2017).

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